MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 10** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason

MATHS MANTHAN | $(5 \times 1 = 5$ Marks)

Q1. A vertical stick 20 cm long casts a shadow 15 cm long on the ground. At the same time, a tower casts a shadow 45 m long on the ground. The height of the tower is: [1]

(A) 40 m (B) 50 m (C) 60 m (D) 75 m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C)

Justification: At the same sun elevation, $\frac{\text{Tower Height}}{45} = \frac{20 \text{ cm}}{15 \text{ cm}} = \frac{4}{3} \implies H = 45 \times \frac{4}{3} = \boxed{60 \text{ m}}$.

Q2. A kite is flying with a string of length 100 m. If the string makes an angle of 30° with the level ground and there is no sag, the height of the kite is: [1]

(A) 50 m (B) $50\sqrt{3}$ m (C) $\frac{100}{\sqrt{3}}$ m (D) 25 m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: $\sin 30^\circ = \frac{h}{100} \implies h = 100 \times \frac{1}{2} = \boxed{50 \text{ m}}$.

Q3. If the length of the shadow of a vertical pole decreases, then the angle of elevation of the sun: [1]

(A) increases (B) decreases (C) remains constant (D) first decreases then increases

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: Since $\tan\theta = \frac{h}{s}$, as shadow s decreases, $\tan\theta$ increases, so the sun's angle of elevation $\boxed{\text{increases}}$.

Q4. A bridge across a river makes an angle of 45° with the river bank. If the length of the bridge across the river is 150 m, the perpendicular width of the river is: [1]

(A) $75\sqrt{2}$ m (B) 75 m (C) $150\sqrt{2}$ m (D) $\dfrac{150}{\sqrt{3}}$ m

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: Perpendicular width $w = 150\sin 45^\circ = \dfrac{150}{\sqrt{2}} = \boxed{75\sqrt{2}\text{ m}}$.

Q5. Assertion (A): An observer of height 1.5 m standing 20 m from a vertical pole sees the top of the pole at an angle of elevation of 45°. The total height of the pole is 21.5 m.
Reason (R): Total height of an object is obtained by adding the height of the observer to the vertical height above eye level: $H = h_{\text{observer}} + d\tan\theta$. [1]

(A) Both (A) and (R) are true; (R) explains (A).
(B) Both are true; (R) does not explain (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: Vertical height above eye $= 20\tan 45^\circ = 20$ m, so total pole height $= 1.5 + 20 = 21.5$ m. Both (A) and (R) are true and (R) explains (A).

SECTION B: Very Short Answer Questions

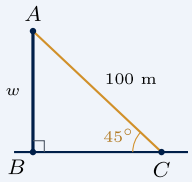
MATHS MANTHAN | $(2 \times 2 = 4$ Marks)

Q6. A bridge over a canal makes an angle of 45° with the canal bank. If the length of the bridge is 100 m, find the perpendicular width of the canal. [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let bridge $AC = 100$ m make $\angle ACB = 45^\circ$ with canal bank BC, where AB is the perpendicular width of the canal.

[0.5 Mark]



$$\sin 45^\circ = \frac{AB}{AC} \implies \frac{1}{\sqrt{2}} = \frac{AB}{100}$$

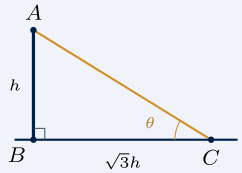
$$\implies AB = \frac{100}{\sqrt{2}} = \boxed{50\sqrt{2}\text{ m} \approx 70.7\text{ m}}$$

Final canal width: [1.5 Marks]

Q7. Find the angle of elevation of the sun when the shadow of a pole h metres high is $\sqrt{3}h$ metres long. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]

Let pole $AB = h$ m cast shadow $BC = \sqrt{3}h$ m at sun elevation $\theta = \angle ACB$. [0.5 Mark]



$$\tan\theta = \frac{AB}{BC} = \frac{h}{\sqrt{3}h} = \frac{1}{\sqrt{3}} = \tan 30^\circ \implies \theta = \boxed{30^\circ}$$

Finding sun elevation: [1.5 Marks]

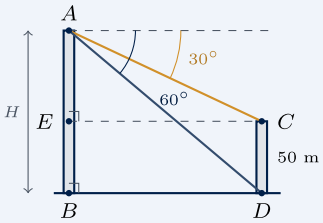
SECTION C: Short Answer Questions

MATHS MANTHAN | $(2 \times 3 = 6$ Marks)

Q8. The angles of depression of the top and bottom of a 50 m high building from the top of a tower are observed to be 30° and 60° respectively. Find the height of the tower and the horizontal distance between them. [3]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let tower $AB = H$ m and 50 m building CD. Draw $CE \perp AB$ so $BE = 50$ m, $AE = H - 50$, $\angle ACE = 30^\circ$, $\angle ADB = 60^\circ$. [0.5 Mark]



In $\triangle ABD$: $BD = CE = \dfrac{H}{\sqrt{3}}$ — (i). In $\triangle AEC$: $CE = \sqrt{3}(H - 50)$ — (ii). [1 Mark]

Equating (i) and (ii): $\dfrac{H}{\sqrt{3}} = \sqrt{3}(H - 50) \implies H = 3H - 150 \implies 2H = 150 \implies H = 75$ m. [1 Mark]

Distance $BD = \dfrac{75}{\sqrt{3}} = 25\sqrt{3} = 25(1.732) = 43.3$ m.

$\therefore$ Height of tower $= \boxed{75 \text{ m}}$, Horizontal distance $= \boxed{25\sqrt{3} \text{ m} \approx 43.3 \text{ m}}$

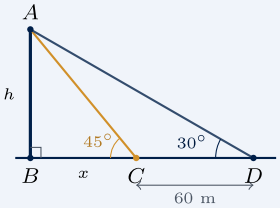
Both required values: [0.5 Mark]

Q9. The shadow of a tower is found to be 60 m shorter when the sun's altitude changes from 30° to 45°. Find the height of the tower. [Take $\sqrt{3} = 1.732$] [3]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let tower $AB = h$ m, shadow at 45° be $BC = x$ m, and shadow at 30° be $BD = (x + 60)$ m ($CD = 60$ m). [0.5 Mark]

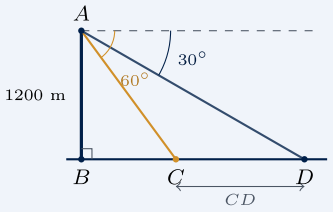
In $\triangle ABC$: $x = h\cot 45^\circ = h$; in $\triangle ABD$: $x + 60 = h\cot 30^\circ = h\sqrt{3}$. [1 Mark]



Substituting $x = h$: $h + 60 = h\sqrt{3} \implies h = \dfrac{60}{\sqrt{3} - 1} = 30(\sqrt{3} + 1) = 30(2.732) = \boxed{81.96 \text{ m}}$. [1.5 Marks]

Q10. An aeroplane flying at an altitude of 1200 m observes two ships on the same side moving away from it in the same vertical plane. The angles of depression of the ships are observed to be 60° and 30° respectively. Find the distance between the two ships. [Take $\sqrt{3} = 1.732$] **[5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]



Let aeroplane be at A at height $AB = 1200$ m, and two ships on the same side be at C (depression 60° $\implies \angle ACB = 60°$) and D (depression 30° $\implies \angle ADB = 30°$). [1 Mark]

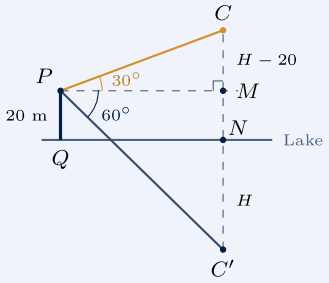
In right $\triangle ABC$: $\tan 60° = \dfrac{1200}{BC} \implies BC = \dfrac{1200}{\sqrt{3}} = 400\sqrt{3}$ m. [1.5 Marks]

In right $\triangle ABD$: $\tan 30° = \dfrac{1200}{BD} \implies BD = 1200\sqrt{3}$ m. [1.5 Marks]

Distance between the two ships $= CD = BD - BC = 1200\sqrt{3} - 400\sqrt{3} = 800\sqrt{3} = 800(1.732) = \boxed{1385.6 \text{ m}}$. [1 Mark]

Q11. The angle of elevation of a stationary cloud from a point 20 m above the surface of a lake is 30° and the angle of depression of its reflection in the lake is 60°. Find the height of the cloud above the surface of the lake. **[5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]



Let observation point P be $PQ = 20$ m above lake QN. Cloud C is at height $CN = H$ m above lake and reflection C' at depth $NC' = H$ m below lake. Let $PM = x$, so $CM = H - 20$ and $C'M = H + 20$. [1 Mark]

In $\triangle PMC$: $\tan 30° = \dfrac{H-20}{x} \implies x = \sqrt{3}(H-20)$ — (i). In $\triangle PMC'$: $\tan 60° = \dfrac{H+20}{x} \implies x = \dfrac{H+20}{\sqrt{3}}$ — (ii). [2 Marks]

Equating (i) and (ii): $\sqrt{3}(H-20) = \dfrac{H+20}{\sqrt{3}} \implies 3(H-20) = H+20.$

$$3H - 60 = H + 20 \implies 2H = 80 \implies \boxed{H = 40 \text{ m}}$$

Solving for cloud height H: [2 Marks]

Q12. CASE STUDY: Residential Garden Avian Flight Dynamics

A migratory bird sits at the top A of mango tree $AB = 10$ m, observed from ground bench O at elevation $45°$. The bird flies away horizontally at constant height 10 m; after 2 seconds, at position C (directly above ground point D), its elevation from O is $30°$.

(i) Find the initial horizontal distance of the tree from the observer (OB). [1]

(ii) Find the horizontal distance of the bird from the observer after 2 seconds (OD). [2]

(iii) Calculate the flight speed of the bird in m/s and km/h. [Take $\sqrt{3} = 1.732$] [2]

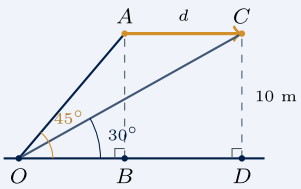
OR

If the bird had instead flown towards the observer until its angle of elevation became $60°$, find its new horizontal distance from the observer. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

(i) In right $\triangle ABO$ ($\angle AOB = 45°, AB = 10$ m):

$$\tan 45° = \frac{10}{OB} \implies 1 = \frac{10}{OB} \implies OB = \boxed{10 \text{ m}}$$



Subpart (i): [1 Mark]

(ii) Let D be the foot of perpendicular from C onto the ground ($CD = 10$ m). In right $\triangle CDO$ ($\angle COD = 30°$):

$$\tan 30° = \frac{10}{OD} \implies OD = \boxed{10\sqrt{3} \text{ m} \approx 17.32 \text{ m}}$$

Subpart (ii): [2 Marks]

(iii) Main Choice: Distance flown in 2 s $= BD = 17.32 - 10 = 7.32$ m. Speed $= \frac{7.32}{2} = \boxed{3.66 \text{ m/s}} = 3.66 \times \frac{18}{5} = \boxed{13.176 \text{ km/h} \approx 13.18 \text{ km/h}}$. [2 Marks]

(iii) OR Choice: If elevation increases to $60°$ while staying at 10 m height, new horizontal distance $= 10 \cot 60° = \frac{10}{\sqrt{3}} = \boxed{\frac{10\sqrt{3}}{3} \text{ m} \approx 5.77 \text{ m}}$. [2 Marks]

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★